

Methods for distance

Principal Coordinates Analysis and related methods

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Introduction

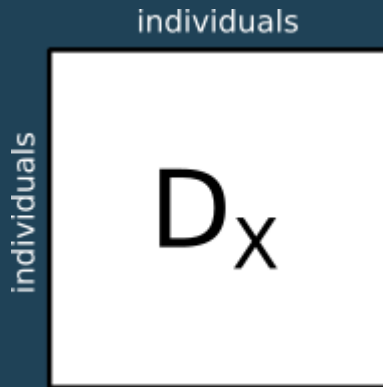
```
library(ade4)
library(adegraphics)
adegpar(paxes.draw = TRUE, pbackground.col = "lightgrey",
        pgrid.col = "white")
data(yanomama)
str(yanomama, max.level = 1)
```

```
## List of 3
## $ geo: num [1:19, 1:19] 0 9 28 152 149 169 172 253 244 82 ...
## $ gen: num [1:19, 1:19] 0 35 44 47 52 57 65 69 34 47 ...
## $ ant: num [1:19, 1:19] 0 96 147 295 284 253 289 507 488 203 ...
```

Differences among 19 Yanomama Indian villages, 3 distance matrices:

- Geographic
- Genetic
- Anthropometric

Introduction

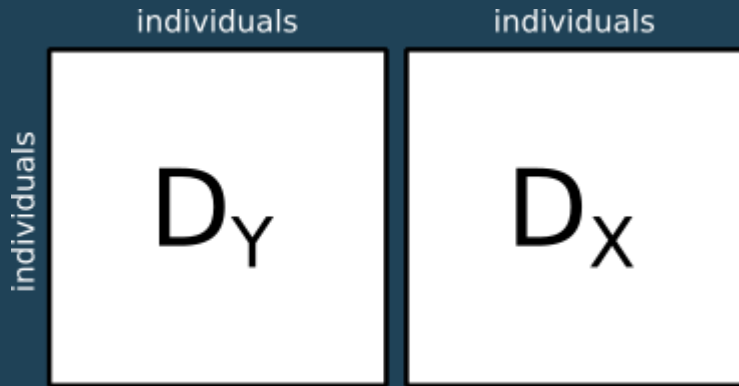


One distance matrix measured on n individuals

Describe the information contained in the table:

- Identify the differences/similarities between individuals

Introduction



Two distance matrices measured on n individuals

Describe the information contained in the two table:

- Identify the differences/similarities between individuals in each table
- Identify relationships between both matrices

Why distances?

- Biological or ecological hypothesis can be defined using distances
- From spatial, genetic, phylogenetic, ecological data, etc.
- Distances can be directly measured
- Distances can be computed from raw data
 - binary
 - numeric
 - percentage
 - factor
 - etc.
- Own definition of differences between individuals.

Dissimilarity and distances

Let E be a set of n individuals. Let x, y, z be 3 elements of E

A **dissimilarity** is an application $d : E \times E \rightarrow \mathbb{R}^+$ satisfying

$$(1) : d(x, y) \geq 0$$

$$(2) : d(x, y) = 0 \Rightarrow x = y$$

$$(3) : d(x, y) = d(y, x)$$

The dissimilarity is **metric** (a.k.a **distance**) if the following condition is satisfied

$$(4) : d(x, y) \leq d(x, z) + d(z, y)$$

It is **ultrametric** if

$$(4) : d(x, y) \leq \max(d(x, z), d(z, y))$$

It is euclidean if we can define n points in an Euclidean space so that distances computed are exactly the dissimilarities.

Methods based on distance matrices

in R, objects of class `dist`

- ultrametric → clustering / classification
- euclidean → Principal Coordinates Analysis (PCoA)
- non-euclidean → Non-Metric Multidimensional Scaling (NMDS)

Principal coordinates analysis

- PCA, CA methods induce implicitly a way to compute distances
- Several other distances have been proposed (e.g., genetic, presence-absence)
- PCoA takes a distance matrix as input and returns coordinates in a low dimensional space that best preserve the original distances.
- 😊 it allows to choose a particular distance measure between sites (or species).
- 😞 it focuses either on individuals or variables, not both.
- Useful if distances are directly recorded or computed from raw data tables

PCoA algorithm

PCoA is based on the diagonalization of a bcentered matrix of squared distances.

$$\mathbf{H} = \left[-\frac{1}{2}d_{ij}^2 - m_i - m_j + m \right]$$

$$\mathbf{H} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^t = \mathbf{U}\mathbf{\Lambda}^{\frac{1}{2}}\mathbf{\Lambda}^{\frac{1}{2}}\mathbf{U}^t = \mathbf{X}\mathbf{X}^t$$

What about non-euclidean distances

```
geo <- as.dist(yanomama$geo)
is.euclid(geo)
```

```
## [1] FALSE
```

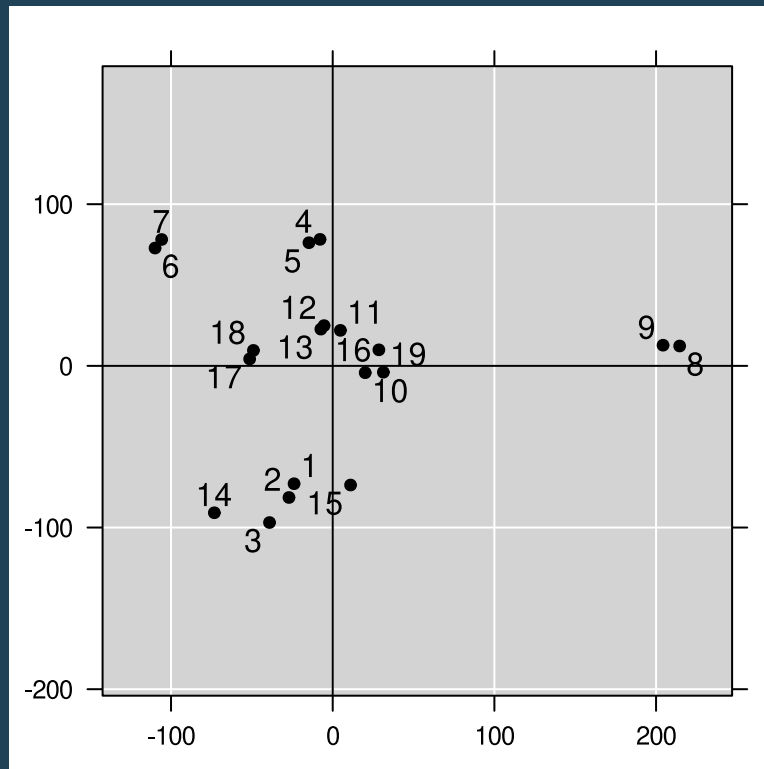
In this case, PCoA would return negative eigenvalues as no euclidean representation is possible to preserve exactly the distances.

Some solutions:

- `cailliez / lingoes` : add constant values to distances
- `quasieucld` : modify distances for quasi-euclidean distances
- NMDS : aims to preserve orders of distances (`vegan::metaMDS`)

PCoA in practice

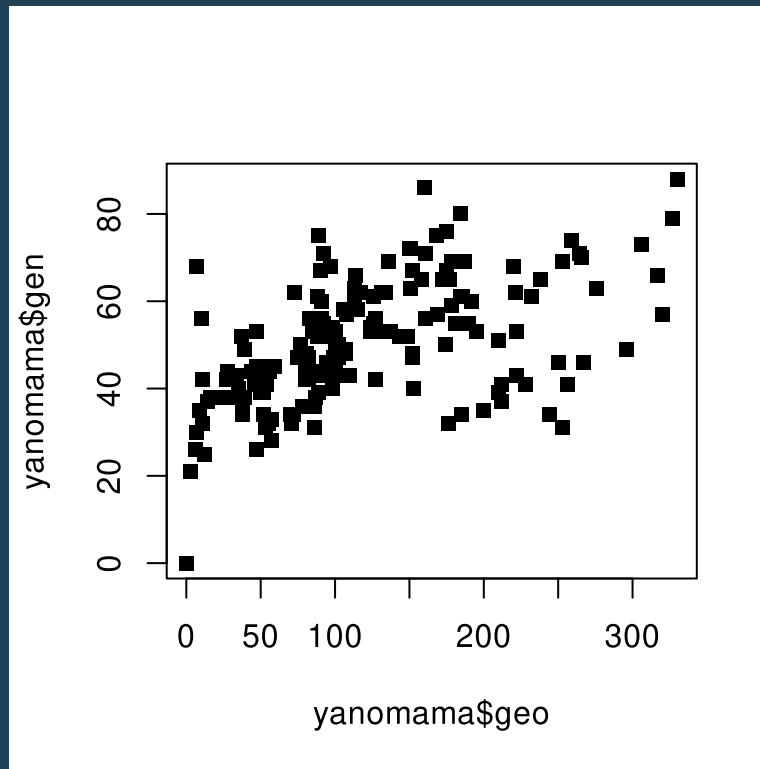
```
geo <- quasieucldid(geo)
pco.geo <- dudi.pco(geo, scan = F)
s.label(pco.geo$li, plabel.optim = TRUE)
```



Relationships among distance matrices

Is there a link between spatial distances and allelic distances ?

```
plot(yanomama$gen ~ yanomama$geo, pch = 15)
```



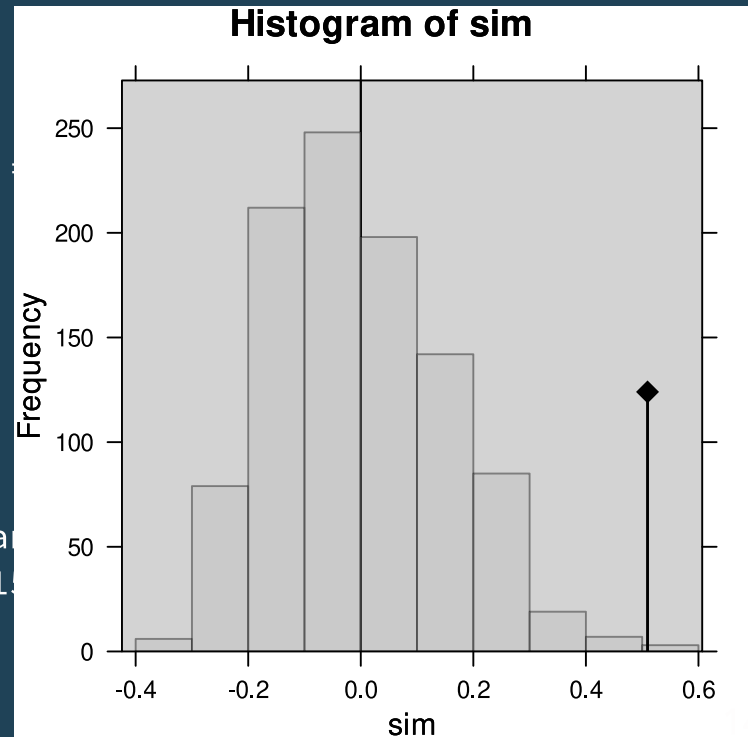
Mantel test

Permutation-based testing procedure based on linear correlation (Pearson's). Can be monotonic (not only linear) if Spearman's correlation is used:

```
gen <- quasieucldid(as.dist(yanoma  
m1 <- mantel.randtest(gen, geo)  
m1
```

```
## Monte-Carlo test  
## Call: mantel.randtest(m1 = gen, m2 = geo)  
##  
## Observation: 0.5095199  
##  
## Based on 999 replicates  
## Simulated p-value: 0.004  
## Alternative hypothesis: greater  
##  
##           Std.Obs  Expectation      Variance  
## 3.301290591 -0.002675429  0.0240711
```

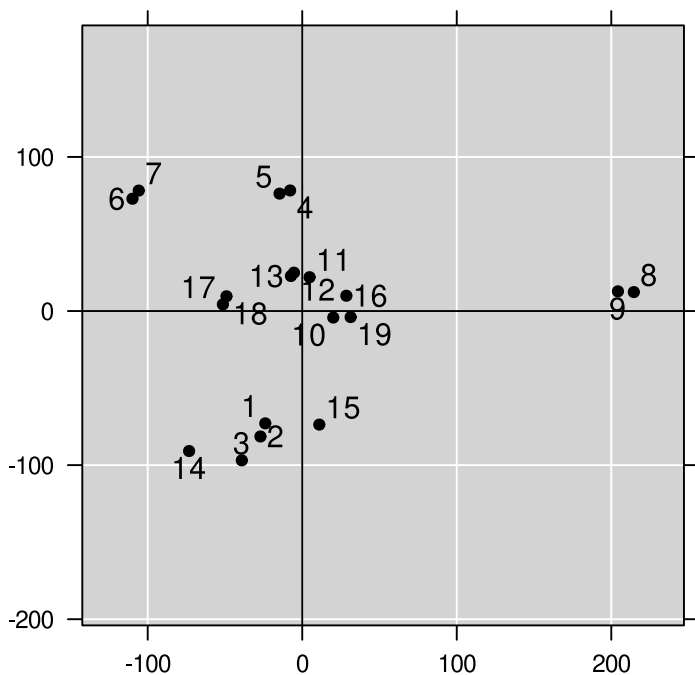
```
plot(m1)
```



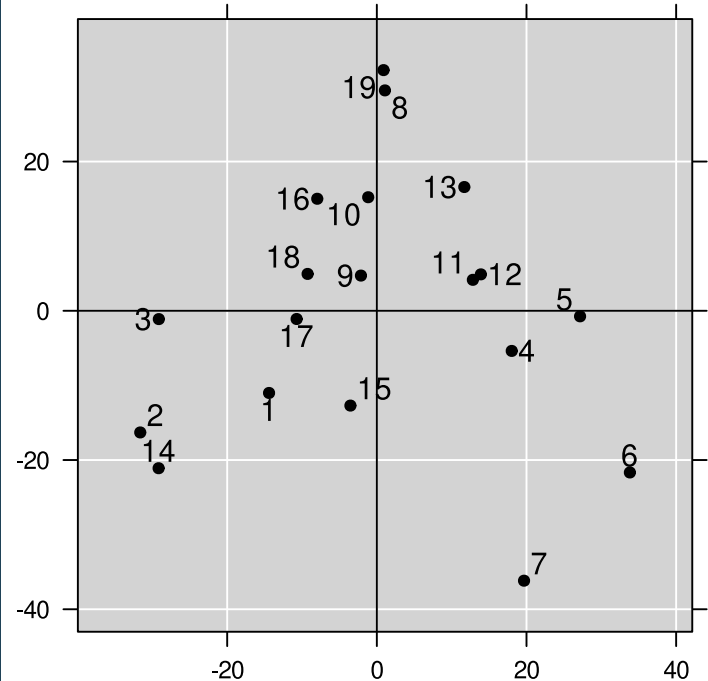
Comparing two configurations

```
pco.gen <- dudi.pco(gen, scan = F)
```

```
s.label(pco.geo$li, plabel.optim
```

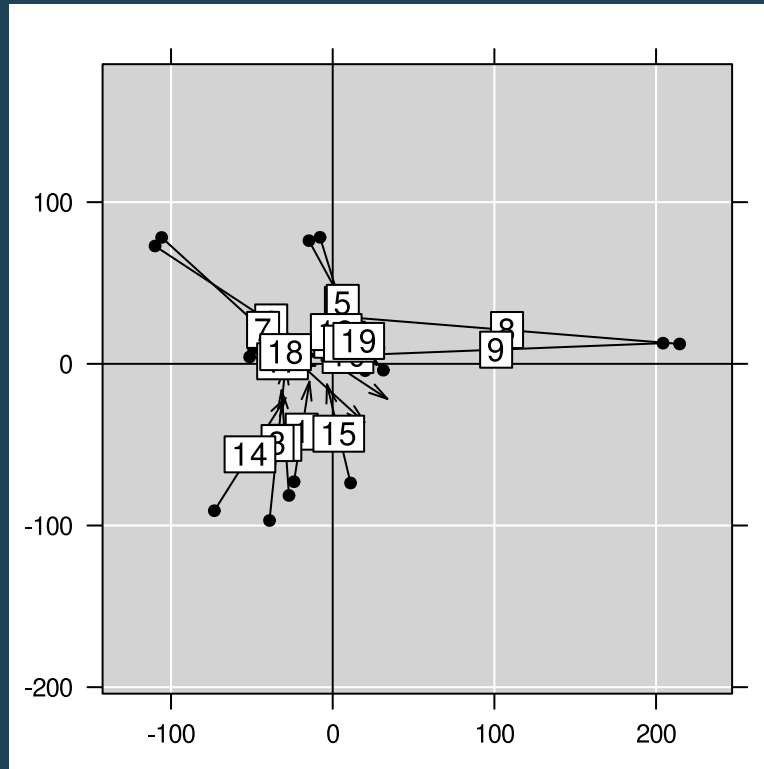


```
s.label(pco.gen$li, plabel.optim
```



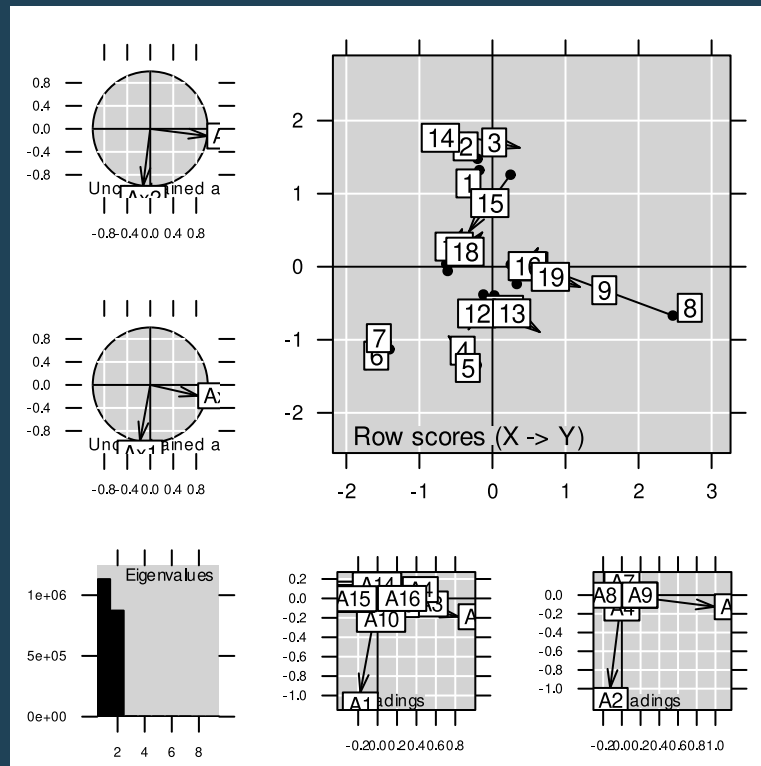
Fit the configurations of points

```
s.match(pco.geo$li, pco.gen$li, plabel.optim = TRUE)
```



Rotate, reflect

```
coi <- coinertia(pco.geo, pco.gen, scannf = FALSE)  
plot(coi)
```



RV coefficient

```
RV.randtest(pco.geo$li, pco.gen$li)
```

```
## Monte-Carlo test
## Call: RV.randtest(df1 = pco.geo$li, df2 = pco.gen$li)
##
## Observation: 0.5128779
##
## Based on 999 replicates
## Simulated p-value: 0.001
## Alternative hypothesis: greater
##
##      Std.Obs Expectation      Variance
## 5.910665783 0.105023185 0.004761438
```

